c78daeb0-49f3-453d-b528-59b7701fe050

<current>

<request>

You are working on an issue in the 'Kong/kongctl' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/kongctl/kongctl (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-06-16T13:35:53.515+00:00</current_datetime>

Consider the following problem statement:

<problem_statement>

This section details on the original issue you should resolve

<issue_title>[e2e-scan] Test: Improve e2e coverage for event-gateway diff child resource types (expand scenario)</issue_title>
<issue_description>### Gap Summary

The `test/e2e/scenarios/event-gateway/diff/scenario.yaml` scenario was introduced to resolve a prior coverage gap, but its testdata config only includes three resource types: `event_gateway`, `event_gateway_backend_cluster`, and `event_gateway_virtual_cluster`. The diff scenario completely omits the child policy resources that every other event-gateway sub-scenario covers:

- `event_gateway_virtual_cluster_produce_policy`
- `event_gateway_virtual_cluster_consume_policy`
- `event_gateway_virtual_cluster_cluster_policy`
- `event_gateway_listener`
- `event_gateway_listener_policy`
- `event_gateway_schema_registry`

````
- `event_gateway_static_key`
- `event_gateway_tls_trust_bundle`
- `event_gateway_dataplane_certificate`

Because `diff` renders its output through a different code path
than the saved-plan `plan` command (see `internal/cmd/root/
verbs/diff/diff.go`), regressions in how child policy fields are
detected or rendered in diff output will go completely
undetected.

### Evidence

- **Diff scenario resource coverage** — run `grep
"resource_type" test/e2e/scenarios/event-gateway/diff/
scenario.yaml | sort | uniq`:
  - Only `event_gateway`, `event_gateway_backend_cluster`,
`event_gateway_virtual_cluster` appear.
- **Diff testdata config** — `test/e2e/scenarios/event-gateway/
diff/testdata/config.yaml` contains no policy or listener blocks
under the virtual cluster.
- **Implementation surface** — `internal/declarative/planner/
egw_control_plane_planner.go` plans all child resources;
`internal/cmd/root/verbs/diff/diff.go` renders the output.
- **Existing child-resource coverage** — each policy type has
its own scenario with full plan/apply coverage but no diff
coverage:
  - `test/e2e/scenarios/event-gateway/produce-policy/
scenario.yaml`
  - `test/e2e/scenarios/event-gateway/consume-policy/
scenario.yaml`
  - `test/e2e/scenarios/event-gateway/cluster-policy/
scenario.yaml`
  - `test/e2e/scenarios/event-gateway/listener/scenario.yaml`
  - `test/e2e/scenarios/event-gateway/listener-policy/
scenario.yaml`

### Proposed Scenario Work

**Expand the existing diff scenario** at `test/e2e/scenarios/
event-gateway/diff/scenario.yaml`.

1. **Expand testdata config** — add a produce policy nested
under the existing `diff-virtual-cluster`:
   ```yaml
   produce_policies:
     - ref: diff-produce-policy
       name: diff-produce-policy
````
````

```
      type: modify_headers
      enabled: true
      config:
        actions:
          - op: set
            key: x-diff-test
            value: initial
```

2. **Update `overlays/003-diff/config.yaml`** — modify the produce policy (e.g., change `config.actions[0].value` or `enabled: false`) to trigger a detected UPDATE.

3. **Add assertions to step `003-diff-with-modifications`** — in the `000-diff-apply-mode` command assert:
   ```yaml
   - select: &gt;-
       changes[?
resource_type=='event_gateway_virtual_cluster_produce_policy'
&amp;&amp;
                resource_ref=='diff-produce-policy'] | [0]
     expect:
       fields:
         action: UPDATE
   ```

4. **Optionally add a sync-mode overlay step** for a policy DELETE to cover sync-mode diff for child resources.

5. **Verify read-only** — confirm Konnect state was NOT modified during the diff step (the existing `001-verify-state-unchanged` pattern).

The change is self-contained within the diff scenario; no new scenario is needed because the diff scenario already owns the EGW + backend cluster + virtual cluster fixture.

Acceptance Criteria

- [ ] `test/e2e/scenarios/event-gateway/diff/testdata/config.yaml` includes at least one child policy resource (produce or consume policy).
- [ ] The existing `overlays/003-diff/config.yaml` overlay modifies the policy to create a detectable diff.
- [ ] Step `003-diff-with-modifications` asserts at least one child resource `UPDATE` appears in the diff output.
- [ ] Konnect state is confirmed unmodified after the diff step

```
(read-only assertion).
- [ ] Scenario passes in CI.

### Reference Patterns

- `test/e2e/scenarios/event-gateway/diff/scenario.yaml` —
existing diff scenario structure to expand
- `test/e2e/scenarios/event-gateway/diff/testdata/config.yaml` —
base testdata to add the policy block to
- `test/e2e/scenarios/event-gateway/diff/overlays/003-diff/
config.yaml` — modify overlay to trigger policy UPDATE
- `test/e2e/scenarios/event-gateway/produce-policy/testdata/
config.yaml` — reference produce-policy config structure









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scanner%22&amp;type=issues">◷</a>
&gt; - [x] expires  on Jul 15, 2026, 6:33 AM UTC




</issue_description>
- Closes Kong/kongctl#1316

## Comments on the Issue (you are @copilot in this section)

<comments>
</comments>
</problem_statement>
```

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.